

Class 10: Structural Bioinformatics Pt. 1

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank (PDB)**

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
head(pdb)
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

Q1. What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this..

Two functions that can help there are `sub()` and `as.numeric()`

```
# We want to get rid of or sub out) commas:
x <- pdb$X.ray
tmp <- sub(",", "", x=pdb$X.ray)
sum( as.numeric(tmp) )
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- sub(",", "", x)
  sum(as.numeric(tmp) )
}
```

```
n.tot <- rm.comma(pdb$Total)
n.xray <- rm.comma(pdb$X.ray)
n.em <- rm.comma(pdb$EM)

n.xray/n.tot * 100
```

```
[1] 80.48577
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

We could also use a different import function `read_csv()` for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```
library(readr)

read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# A tibble: 6 x 9
```

	`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1	Protein (only)	180758	23111	12813	348	229	84
2	Protein/Oligosacch~	10488	3741	34	8	11	1
3	Protein/NA	9205	6751	287	26	8	0
4	Nucleic acid (only)	3154	250	1578	3	15	3
5	Other	178	27	35	4	0	0
6	Oligosaccharide (o~	11	0	6	0	1	0

```
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
n.tot <- pdb$Total
n.xray <- sum(pdb$`X-ray`)
```

How many total protein structures are there in the dataset?

```
pdb$Total[1]
```

```
[1] "217,375"
```

The total number of protein sequences in UniProt is 202,556,314.

217375/202556314 * 100

[1] 0.1073158

Key-point: We have very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method X-ray crystallography).

Q2: What proportion of structures in the PDB are protein?

Approximately ~85–90% of structures in the PDB are proteins.

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are approximately 1227 HIV-1 protease structures in the PDB.

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at <https://molstar.org/viewer/>



Figure 1: The HIV-protease enzyme is a homodimer with two chains

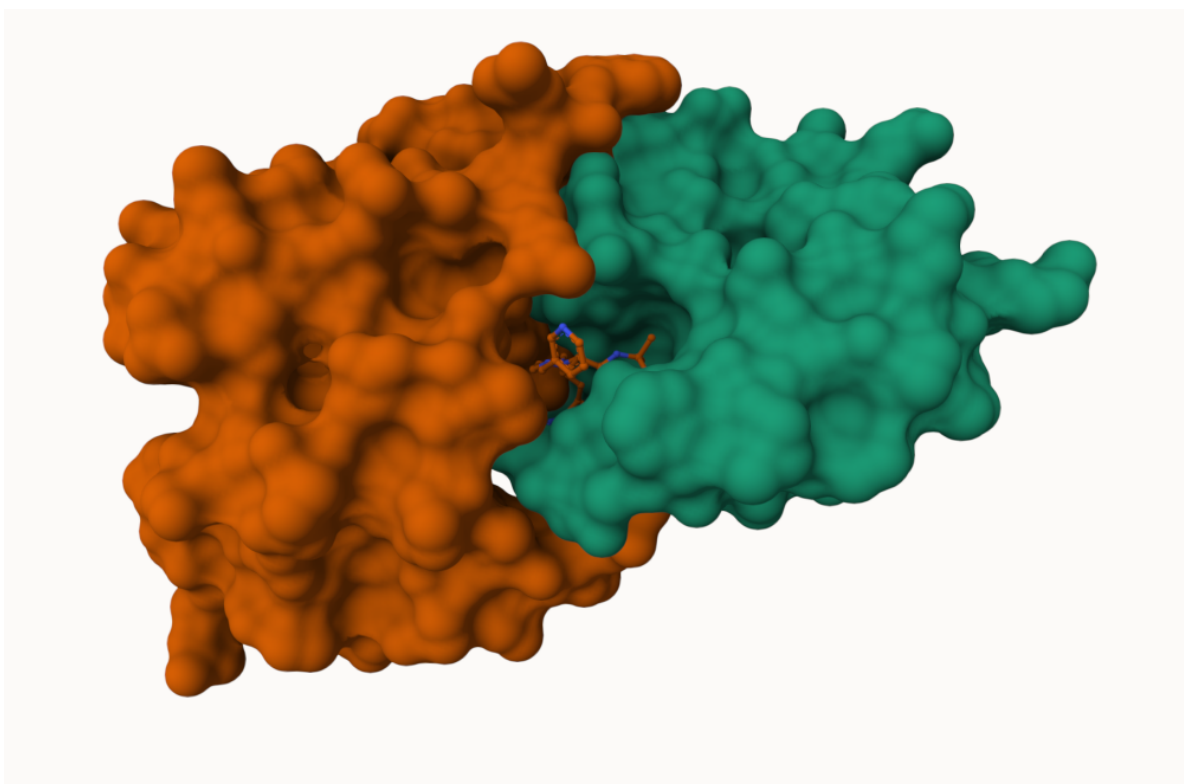


Figure 2: Surface display showing the binding cleft site for the inhibitor (drug) molecule

```
#Pak can install from multiple sources
#install.packages("pak")

pak::pkg_install( c("bioboot/bio3dview", #Github
                    "NGLVieweR",         #CRAN
                    "bioc::msa"))         #Bioconductor
```

! Using bundled GitHub PAT. Please add your own PAT using `gitcreds::gitcreds\_set()`.

Loading metadata database

Loading metadata database ... done

No downloads are needed

3 pkgs + 47 deps: kept 44 [6.8s]

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Water molecules normally have three atoms (H O), but in X-ray crystal structures only the oxygen atom is visible. This is because hydrogen atoms are too small and have too few electrons to be detected reliably, so only the oxygen atom is resolved in the structure.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

A conserved water molecule (HOH) is located in the binding site, forming hydrogen bonds between the ligand and the catalytic residues. The residue number corresponds to a labeled HOH molecule in the active site (e.g., HOH 301).

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

HIV protease has flexible “flap” regions that open and close over the active site. This flexibility allows large ligands such as indinavir to enter the binding pocket. Once inside, the flaps close to stabilize binding through interactions with the ligand and catalytic residues.

Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

HIV protease is a homodimer, and the dimer interface forms β -sheet interactions between the two chains. These inter-chain β -sheets stabilize the structure and are not present in a single monomer, meaning they only form upon dimerization.

Getting Started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d)
```

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD  
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE  
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP  
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198 amino acid residues

Q8: Name one of the two non-protein residues?

Q8: Name one non-protein residue

Non-protein/nucleic resid values: [HOH (127), MK1 (1)] ; HOH (water) OR MK1 (ligand)

Q9: How many protein chains are in this structure?

Q9: How many protein chains?

2 (values: A B)

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom, 6)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

There are lots of functions that can work with these `pdb` objects

```
head( pdbseq(pdb) )
```

```
  1    2    3    4    5    6  
"P" "Q" "I" "T" "L" "W"
```

We can have a quick interactive view of any of these `pdb` objects:

```
#library(bio3dview)

#view.pdb(pdb)
```

Let's try a custom view

```
#view.pdb(pdb, colorScheme = "sse", backgroundColor = "black")
```

Q. Create a custom view highlighting the active site ASP (**resno=25**), the two chains (in your choice of colors) and the ligand all on a custom color background?

```
#library(NGLViewer)
#active.site <- atom.select(pdb, resno=25)

#view.pdb(pdb,
  #cols = c("purple", "orange"),
  # highlight = active.site,
  # highlight.style = "spacefill",
  # backgroundColor = "lightpink") |>
# setRock()
```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given **pdb** object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, **rm.alt=TRUE**

A quick structure summary

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

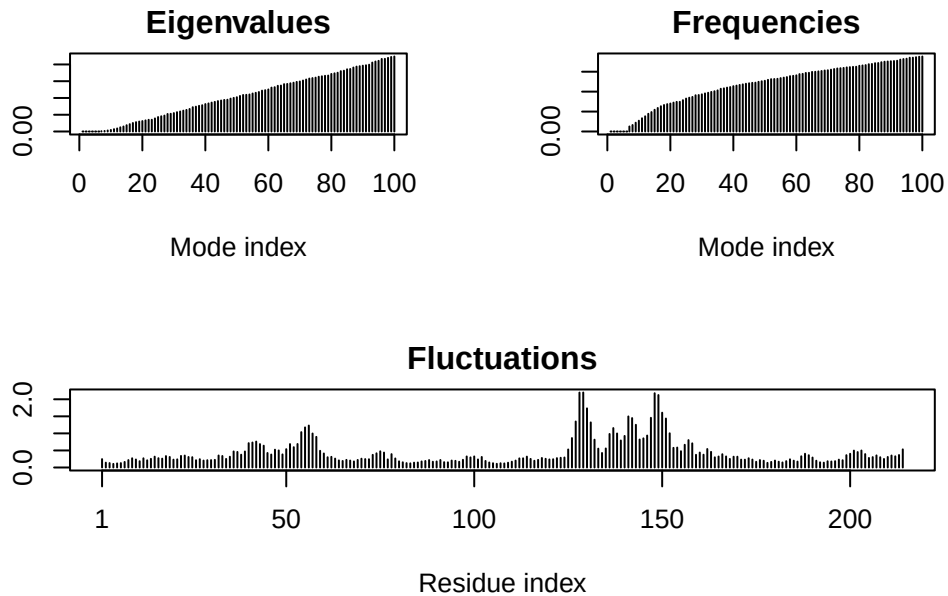
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

```
m <- nma(adk)
```

Building Hessian... Done in 0.03 seconds.
Diagonalizing Hessian... Done in 0.33 seconds.

```
plot(m)
```



View the results with an

```
#view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

Our first step is find a sequence for this family. We will use the database ID “1ake_A” here:

```
id <- "1ake_A"  
aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```
      1      .      .      .      .      .      60  
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
      1      .      .      .      .      .      60  
  
     61      .      .      .      .      .      120  
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI  
     61      .      .      .      .      .      120  
  
    121      .      .      .      .      .      180  
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
    121      .      .      .      .      .      180  
  
    181      .      .      .      214  
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG  
    181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

fasta

Alignment dimensions:

1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = ZS38K9JR014

.....

Reporting 96 hits

```
head(blast$hit.tbl)
```

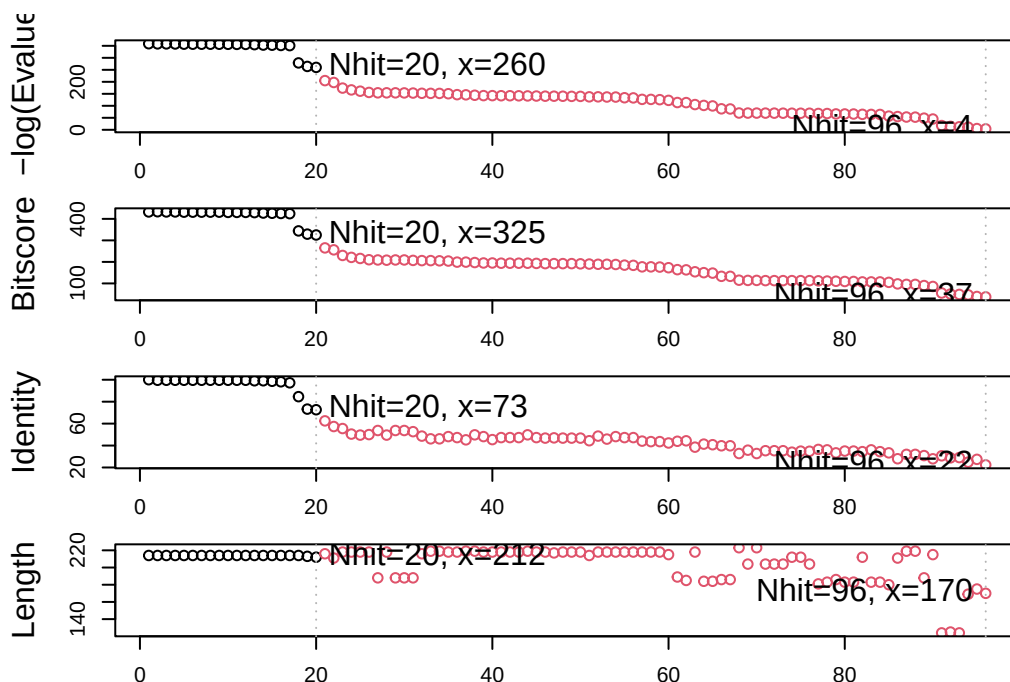
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_1779385	1AKE_A	100.000	214	0	0	1
2	Query_1779385	8BQF_A	99.533	214	1	0	1
3	Query_1779385	4X8M_A	99.533	214	1	0	1
4	Query_1779385	6S36_A	99.533	214	1	0	1
5	Query_1779385	9R6U_A	99.533	214	1	0	1
6	Query_1779385	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evalue	bitscore	positives	mlog.evalue	pdb.id	acc
1	214	1	214	1.81e-156	432	100.00	358.6099	1AKE_A	1AKE_A
2	214	21	234	2.97e-156	433	100.00	358.1147	8BQF_A	8BQF_A
3	214	1	214	3.25e-156	432	100.00	358.0246	4X8M_A	4X8M_A
4	214	1	214	4.76e-156	432	100.00	357.6430	6S36_A	6S36_A
5	214	1	214	1.06e-155	431	99.53	356.8424	9R6U_A	9R6U_A
6	214	1	214	1.25e-155	431	99.53	356.6775	9R71_A	9R71_A

```
hits <- plot(blast)
```

* Possible cutoff values: 260 3
Yielding Nhits: 20 96

* Chosen cutoff value of: 260
Yielding Nhits: 20



```
hits$ pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$ pdb.id, path="pdbc", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/1AKE.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/8BQF.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/4X8M.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/6S36.pdb exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb exists. Skipping download

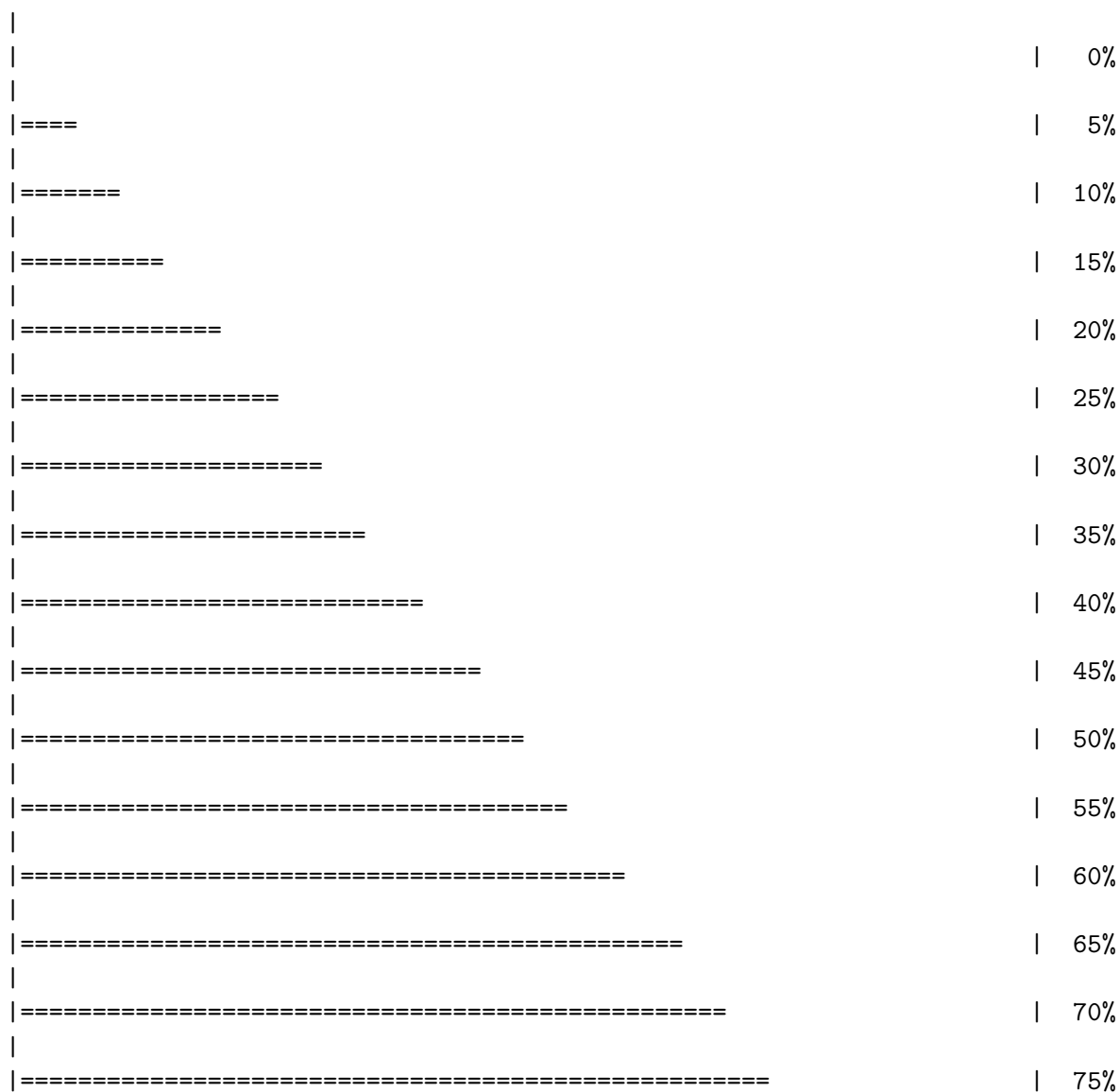
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb exists. Skipping download



	=====	80%
	=====	85%
	=====	90%
	=====	95%
	=====	100%

Align and superpose all these ADK structures

```
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb
```

```
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
```

```

.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/8BQF_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbc/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/9R6U_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbc/split_chain/9R71_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/8Q2B_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbc/split_chain/8RJ9_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbc/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbc/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbc/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbc/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbc/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbc/split_chain/8PVW_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbc/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE

```

pdb/seq: 20 name: pdbs/split_chain/4NP6_A.pdb

pdbs

	1	.	.	.	40
[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:5] 9R6U_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:6] 9R71_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:7] 8Q2B_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:8] 8RJ9_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:9] 6RZE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:10] 4X8H_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:11] 3HPR_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:12] 1E4V_A.pdb	--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:13] 5EJE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:14] 1E4Y_A.pdb	--MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:15] 3X2S_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:16] 6HAP_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:17] 6HAM_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:18] 8PVW_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDM	LRAA			
[Truncated_Name:19] 4K46_A.pdb	--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDM	LRAA			
[Truncated_Name:20] 4NP6_A.pdb	NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDM	LRAA			
	***** ***** *~*****				
	1	.	.	.	40
	41	.	.	.	80
[Truncated_Name:1] 1AKE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:2] 8BQF_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---				
[Truncated_Name:3] 4X8M_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:4] 6S36_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:5] 9R6U_A.pdb	VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:6] 9R71_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:7] 8Q2B_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:8] 8RJ9_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:9] 6RZE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:10] 4X8H_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:11] 3HPR_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:12] 1E4V_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:13] 5EJE_A.pdb	VKSGSELGKQAKDIMDACKLVDELVIALVKERIAQEDCR				

[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR
[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE

^* *^*** *** ^~* **~*~~~~~*~^*** *
 41 . . . 80

[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	81	120
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD		
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD		
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVGVVVDYVIEFDVADSVIVE		
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADDVIVE		

*****^~** *^ *****^***** * ^**^
 81 . . . 120

[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	121	160
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKA		
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		

[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRRAHLPSGRTYHVNVNPPKVEGKDDVTGEDLVIRED
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG

[Truncated_Name:4] 6S36_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:5] 9R6U_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:6] 9R71_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:7] 8Q2B_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:8] 8RJ9_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:9] 6RZE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:10] 4X8H_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:11] 3HPR_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:12] 1E4V_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:13] 5EJE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:14] 1E4Y_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb	TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb	TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb	TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb	TKQVSEVSADIAKALA

** * ** *~* *
 201 . 216

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

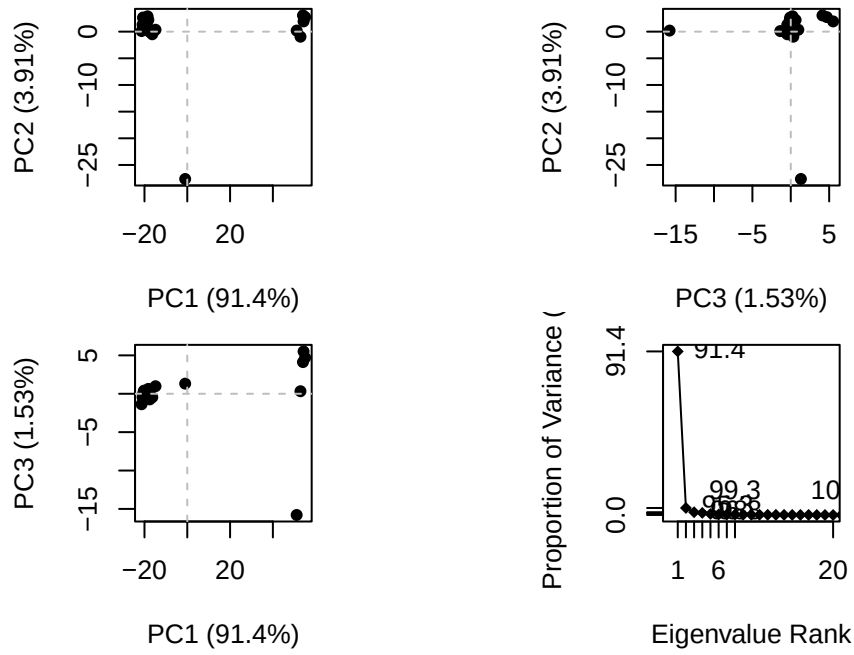
```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structural view

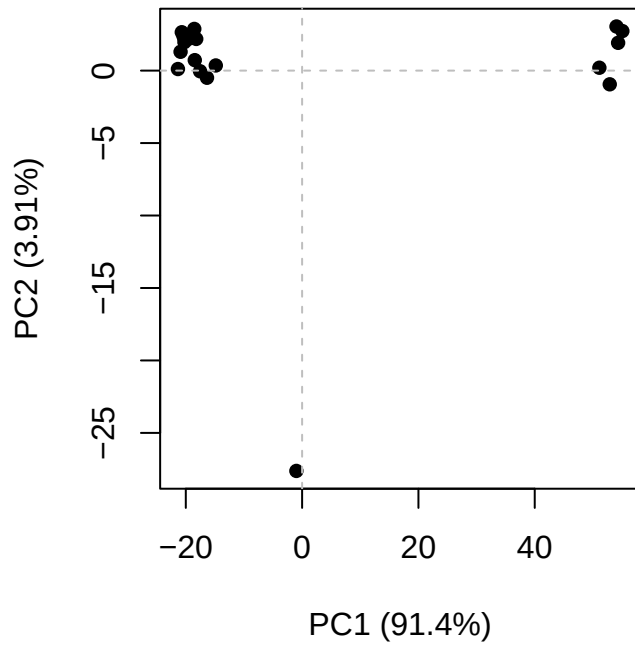
```
#view.pdbs(pdbs)
```

PCA of all this structural data (x, y and z atom coordinates):

```
pc <- pca(pdbs)
plot(pc)
```



```
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

```
#view.pca(pc)
```

```
mktrj(pc, file="pca.pdb")
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q11. Which of the above packages is not found on BioConductor or CRAN?

bio3dview

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

YES — pak can install from:

CRAN GitHub Bitbucket